SIMATSENGINEERING

AI-Powered Smart Water Reservoir Management and Demand Forecasting Platform

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1. Introduction

Water reservoirs are critical infrastructure assets that supply drinking water, support irrigation for agriculture, and meet industrial water demand. Increasing climate variability, unpredictable rainfall patterns, and growing population pressure make efficient reservoir management a significant challenge. Traditional reservoir management approaches rely heavily on manual observation and historical experience, which often results in inefficient allocation of water resources, delayed responses to drought conditions, and avoidable water shortages.

This document presents a comprehensive Test Case Design for an AI-Powered Smart Water Reservoir Management and Demand Forecasting Platform. The platform leverages Artificial Intelligence, Internet of Things (IoT) sensors, satellite monitoring, and cloud analytics to monitor reservoir conditions in real time, forecast water demand, predict future water availability, and recommend optimal distribution strategies. The purpose of this document is to define a structured, traceable, and technique-driven suite of test cases that verify the correctness, functionality, reliability, security, and usability of the proposed system prior to deployment.

2. Problem Statement

Traditional reservoir management relies on historical observations and manual decision-making, often leading to inefficient water allocation during droughts or heavy rainfall. An AI-powered reservoir management platform should monitor reservoir levels, rainfall forecasts, inflow and outflow rates, and water demand patterns. The system should predict future water availability, recommend optimal distribution strategies, and support authorities in sustainable water resource planning.

Key Objectives of the System:

- Monitor reservoir water levels.
- Forecast water demand.
- Predict water availability.
- Optimize water distribution.
- Analyze rainfall patterns.
- Generate water management reports.
- Support drought planning.
- Improve water resource utilization.

Expected Outcomes:

- Efficient water allocation.
- Improved reservoir management.
- Reduced water shortages.
- Better drought preparedness.
- Sustainable water utilization.
- Accurate demand forecasting.
- Enhanced environmental conservation.
- Improved public water supply.

3. Scope of Testing

The scope of this test case design covers functional testing of all eight core system objectives (reservoir monitoring, demand forecasting, availability prediction, distribution optimization, rainfall analysis, report generation, drought planning, and resource utilization), together with non-functional testing of performance, reliability, security, and usability. Testing covers normal (positive), boundary, invalid (negative), and exceptional conditions for each module, using formally recognized test design techniques including Equivalence Partitioning (EP), Boundary Value Analysis (BVA), Decision Table Testing, and State Transition Testing. Integration between the AI forecasting engine and external data sources (IoT sensors, satellite feeds, weather APIs) is included; testing of the underlying machine-learning model training pipeline is out of scope for this document and is assumed to be covered separately under model validation activities.

4. Functional Requirements

The following functional requirements (FR) were identified from the problem statement and objectives, and form the basis for test scenario and test case derivation.

Req. ID	Requirement Description
FR-01	The system shall continuously monitor real-time reservoir water levels using IoT sensors and satellite data feeds.
FR-02	The system shall record and store historical reservoir level data at configurable time intervals (e.g., hourly, daily).
FR-03	The system shall forecast water demand for drinking, irrigation, and industrial use based on historical consumption patterns and seasonal trends.
FR-04	The system shall predict future water availability using inflow, outflow, and rainfall forecast data.
FR-05	The system shall recommend optimal water distribution strategies across drinking, irrigation, and industrial sectors.
FR-06	The system shall analyze rainfall patterns using satellite and weather station data to support inflow prediction.
FR-07	The system shall generate periodic water management reports (daily, weekly, monthly) summarizing levels, demand, and distribution.
FR-08	The system shall raise drought-risk alerts when predicted water availability falls below defined safety thresholds.
FR-09	The system shall allow authorized users to configure reservoir capacity, safe minimum level, and alert thresholds.
FR-10	The system shall support role-based access for administrators, water resource officers, and public viewers.
FR-11	The system shall allow export of reports in PDF and CSV formats.
FR-12	The system shall display reservoir status and forecasts on a dashboard with graphical visualizations.
FR-13	The system shall send automated notifications (SMS/email/app) to designated authorities when critical thresholds are breached.
FR-14	The system shall allow manual override/correction of sensor readings by authorized officers with audit logging.
FR-15	The system shall integrate with external weather APIs to fetch rainfall forecast data periodically.

5. Non-Functional Requirements

The following non-functional requirements (NFR) address performance, reliability, security, usability, and scalability attributes of the system.

Req. ID	Requirement Description
NFR-01	The system shall process and update reservoir level data within 5 seconds of sensor data receipt.
NFR-02	The system shall maintain 99.5% uptime for dashboard and alerting services.
NFR-03	The system shall support at least 500 concurrent dashboard users without performance degradation.
NFR-04	The system shall encrypt all sensor and user data in transit and at rest.
NFR-05	The system shall be usable by non-technical water resource officers with minimal training (usability).
NFR-06	The system shall retain historical data for a minimum of 10 years for trend analysis and audit purposes.
NFR-07	The system shall be scalable to accommodate additional reservoirs without architectural redesign.
NFR-08	The system shall log all critical actions (overrides, configuration changes) for audit and traceability.

6. Test Scenarios

Test scenarios were derived module-by-module from the functional and non-functional requirements above. Each scenario is later expanded into one or more detailed test cases in Section 8.

6.1 Reservoir Water Level Monitoring (RLM)

Related Requirements: FR-01, FR-02, FR-09, NFR-01

- Verify real-time reservoir level is displayed correctly on the dashboard
- Verify historical level data is stored at configured intervals
- Verify system behavior when sensor readings are at reservoir minimum/maximum capacity
- Verify system handles invalid or out-of-range sensor readings
- Verify system behavior on sensor communication failure
- Verify authorized officer can manually correct a sensor reading with audit log entry

6.2 Water Demand Forecasting (WDF)

Related Requirements: FR-03, NFR-05

- Verify demand forecast is generated using historical consumption data
- Verify forecast adjusts for seasonal variation (e.g., summer irrigation peak)
- Verify system handles insufficient historical data gracefully
- Verify forecast for multiple sectors (drinking, irrigation, industrial) is generated independently
- Verify forecast output is rejected for invalid date range input

6.3 Water Availability Prediction (WAP)

Related Requirements: FR-04, FR-06

- Verify availability prediction combines inflow, outflow, and rainfall forecast data
- Verify prediction updates when rainfall forecast data changes significantly
- Verify system flags prediction as unreliable when input data sources are unavailable
- Verify prediction correctly handles zero inflow scenario

6.4 Water Distribution Optimization (WDO)

Related Requirements: FR-05

- Verify system recommends distribution allocation across sectors based on priority rules
- Verify system re-optimizes allocation when predicted availability decreases
- Verify system applies drinking-water priority rule during scarcity (decision table)
- Verify system rejects an allocation plan that exceeds available water

6.5 Rainfall Pattern Analysis (RPA)

Related Requirements: FR-06, FR-15

- Verify rainfall data is fetched successfully from external weather API
- Verify system handles malformed data returned by weather API
- Verify rainfall trend analysis correctly identifies anomalies vs. historical averages

6.6 Water Management Report Generation (WMR)

Related Requirements: FR-07, FR-11

- Verify daily/weekly/monthly reports are generated with correct data aggregation
- Verify report export in PDF and CSV formats
- Verify report generation for a reservoir with no data in the selected period

6.7 Drought Planning Support (DPS)

Related Requirements: FR-08, FR-13

- Verify drought-risk alert triggers when predicted availability falls below threshold
- Verify multi-channel notification (SMS/email/app) is sent to designated authorities
- Verify escalation when drought condition persists beyond configured duration

6.8 Security, Access Control and Auditability (SEC)

Related Requirements: FR-10, FR-14, NFR-04, NFR-08

- Verify role-based access restricts feature availability per user role
- Verify invalid login attempts are handled securely
- Verify sensitive data is encrypted in transit
- Verify all critical actions are captured in the audit log

6.9 Performance, Reliability and Usability (PERF)

Related Requirements: NFR-01, NFR-02, NFR-03, NFR-05, NFR-07

- Verify sensor data update latency meets the 5-second requirement
- Verify dashboard performance under concurrent user load
- Verify system remains available during a single reservoir node failure
- Verify a first-time user can complete a core task without training

7. Test Design Techniques Applied

Multiple recognized test design techniques were applied across the modules to ensure systematic and efficient test coverage:

7.1 Equivalence Partitioning (EP)

Input domains such as reservoir level (0-100%), forecast horizon (days), and user roles were divided into valid and invalid partitions. One representative value from each partition was selected for testing, e.g., reservoir level partitions: {invalid negative}, {valid 0-100}, {invalid >100}.

7.2 Boundary Value Analysis (BVA)

Boundaries of critical thresholds were specifically targeted, since defects frequently occur at boundary values. Examples include reservoir level at 0% and 100%, drought threshold at exactly 25%, forecast horizon at 365/366 days, and login lockout at the 5th/6th failed attempt.

7.3 Decision Table Testing

The Water Distribution Optimization module involves multiple interacting conditions (available water vs. sector demands, drinking-water priority rule). A decision table was used to enumerate combinations of conditions and their corresponding allocation outcomes, ensuring all business rule combinations are exercised.

Condition / Action	Rule 1	Rule 2	Rule 3	Rule 4
Available >= Total Demand	Y	N	N	N
Available >= Drinking Demand	-	Y	N	N
Available > 0	-	-	Y	N
Action: Allocate full demand to all sectors	X			
Action: Allocate 100% drinking, proportion remainder		X		
Action: Allocate all to drinking, zero to others + alert			X	
Action: Trigger critical drought emergency protocol				X

7.4 State Transition Testing

The Drought Planning Support module exhibits distinct states (Normal → Drought Watch → Severe Drought → Normal). State transition testing was used to verify that the system moves correctly between these states based on predicted water availability and elapsed time, including boundary and invalid transitions.

Current State	Trigger Event	Next State
Normal	Predicted availability falls below drought threshold (e.g., <25%)	Drought Watch

Drought Watch	Predicted availability recovers above threshold	Normal
Drought Watch	Drought condition persists beyond 7 consecutive days	Severe Drought
Severe Drought	Predicted availability recovers above threshold	Normal
Normal	Predicted availability remains above threshold	Normal (no change)

8. Detailed Test Cases

This section presents detailed test cases for each module, covering normal, boundary, invalid, and exceptional conditions. Each test case is uniquely identified, mapped to a test design technique, and includes Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status. Actual Result and Pass/Fail Status below reflect execution against a correctly functioning build during a sample dry-run and would be updated during formal test execution.

8.1 Reservoir Water Level Monitoring (RLM)

Applicable Test Design Technique(s): Boundary Value Analysis, Equivalence Partitioning

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-RLM-01	Display current reservoir level within normal operating range	1. Login as Water Resource Officer. 2. Navigate to Dashboard. 3. Observe current reservoir level widget.	Sensor input = 65% of capacity (Range: 0-100%)	Dashboard displays 65% with corresponding graphical gauge and green (safe) status indicator.	As Expected	Pass
TC-RLM-02	Reservoir level at minimum boundary (0%)	1. Simulate sensor input at 0% capacity. 2. Observe dashboard and alert status.	Sensor input = 0%	System displays 0% level, triggers 'Critical Low' alert and red status indicator.	As Expected	Pass
TC-RLM-03	Reservoir level at maximum boundary (100%)	1. Simulate sensor input at 100% capacity. 2. Observe dashboard and alert status.	Sensor input = 100%	System displays 100% level, triggers 'Overflow Risk' alert.	As Expected	Pass
TC-RLM-04	Reservoir level just below minimum safe threshold (boundary -1)	1. Configure safe minimum threshold = 20%. 2. Simulate sensor input = 19%. 3. Observe alert.	Threshold = 20%, Sensor input = 19%	System raises low-level warning as reading is below configured threshold.	As Expected	Pass
TC-RLM-05	Reservoir level exactly at minimum safe threshold (boundary)	1. Configure safe minimum threshold = 20%. 2. Simulate sensor input = 20%. 3. Observe alert.	Threshold = 20%, Sensor input = 20%	System treats level as within safe range; no warning raised (boundary inclusive).	As Expected	Pass
TC-RLM-06	Invalid sensor reading (negative value)	1. Simulate sensor input = -5%. 2. Observe system validation response.	Sensor input = -5%	System rejects the reading, logs a sensor-fault error, and retains last valid reading on dashboard.	As Expected	Pass

TC-RLM-07	Invalid sensor reading (value exceeding 100%)	1. Simulate sensor input = 130%. 2. Observe system validation response.	Sensor input = 130%	System rejects reading as out-of-range and flags sensor malfunction.	As Expected	Pass
TC-RLM-08	Sensor communication timeout	1. Disconnect sensor feed. 2. Wait beyond configured timeout (30s). 3. Observe dashboard.	No data received for 30+ seconds	Dashboard shows 'Sensor Offline' status and notifies maintenance team automatically.	As Expected	Pass
TC-RLM-09	Manual correction of erroneous sensor reading by authorized officer	1. Login as authorized officer. 2. Select flawed reading. 3. Enter corrected value with justification. 4. Submit.	Original = 45%, Corrected = 52%, Reason = 'Sensor drift'	Corrected value updates the record; an audit log entry is created capturing user, old/new value, timestamp, and reason.	As Expected	Pass
TC-RLM-10	Unauthorized user attempts manual correction	1. Login as Public Viewer role. 2. Attempt to access reading correction feature.	Role = Public Viewer	System denies access and displays 'Insufficient Privileges' message.	As Expected	Pass

8.2 Water Demand Forecasting (WDF)

Applicable Test Design Technique(s): Equivalence Partitioning, Decision Table

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-WDF-01	Generate demand forecast with sufficient historical data (12 months)	1. Select reservoir. 2. Select forecast horizon = 30 days. 3. Click 'Generate Forecast'.	Historical data = 12 months, Horizon = 30 days	System generates a 30-day demand forecast graph with sector-wise breakdown (drinking/irrigation/industrial).	As Expected	Pass
TC-WDF-02	Forecast reflects seasonal irrigation peak in summer months	1. Select forecast period covering April-June. 2. Generate forecast. 3. Compare with baseline demand.	Period = Apr-Jun, historical seasonal pattern present	Forecasted irrigation demand shows expected upward trend consistent with historical summer usage.	As Expected	Pass

TC-WDF-03	Forecast attempt with insufficient historical data (< 30 days)	1. Select newly added reservoir with only 10 days of data. 2. Attempt to generate forecast.	Historical data available = 10 days	System displays a warning: 'Insufficient historical data for reliable forecast' and disables forecast generation or shows low-confidence indicator.	As Expected	Pass
TC-WDF-04	Forecast for each demand sector generated independently	1. Generate forecast. 2. Verify drinking, irrigation, and industrial demand figures are displayed separately.	Sectors = Drinking, Irrigation, Industrial	Each sector's forecast is computed and displayed independently with its own confidence interval.	As Expected	Pass
TC-WDF-05	Invalid forecast horizon (0 days)	1. Enter forecast horizon = 0. 2. Click 'Generate Forecast'.	Horizon = 0	System displays validation error: 'Forecast horizon must be at least 1 day'.	As Expected	Pass
TC-WDF-06	Invalid forecast horizon (negative value)	1. Enter forecast horizon = -10. 2. Click 'Generate Forecast'.	Horizon = -10	System rejects input with validation error and does not call the forecasting engine.	As Expected	Pass
TC-WDF-07	Forecast horizon at maximum supported boundary (365 days)	1. Enter forecast horizon = 365. 2. Generate forecast.	Horizon = 365 (maximum allowed)	System successfully generates a long-term forecast, flagged as 'lower confidence' due to extended horizon.	As Expected	Pass
TC-WDF-08	Forecast horizon exceeding maximum boundary (366 days)	1. Enter forecast horizon = 366. 2. Click 'Generate Forecast'.	Horizon = 366	System rejects request: 'Maximum forecast horizon is 365 days'.	As Expected	Pass
TC-WDF-09	Invalid date range where end date precedes start date	1. Set start date = 2026-09-01. 2. Set end date = 2026-08-01. 3. Generate forecast.	Start = 2026-09-01, End = 2026-08-01	System displays error: 'End date must be after start date' and blocks forecast generation.	As Expected	Pass

8.3 Water Availability Prediction (WAP)

Applicable Test Design Technique(s): Equivalence Partitioning, State Transition Testing

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-WAP-01	Generate water availability prediction with all data sources available	1. Ensure inflow, outflow, and rainfall data are current. 2. Trigger prediction engine.	Inflow = 500 ML/day, Outflow = 300 ML/day, Rainfall forecast = Moderate	System computes and displays projected water availability for next 7/30/90 days with confidence score.	As Expected	Pass
TC-WAP-02	Prediction updates after significant rainfall forecast revision	1. Generate initial prediction. 2. Update rainfall forecast API data to reflect heavy rainfall. 3. Re-trigger prediction.	Rainfall forecast revised from 'Low' to 'Heavy'	Predicted availability increases proportionally and dashboard reflects updated projection with timestamp.	As Expected	Pass
TC-WAP-03	Rainfall API data source unavailable	1. Simulate rainfall API outage. 2. Trigger prediction.	Rainfall API = unreachable	System generates prediction using last known rainfall data and flags result as 'Reduced Confidence - Rainfall data stale'.	As Expected	Pass
TC-WAP-04	Both inflow and rainfall data sources unavailable	1. Simulate outage of inflow sensors and rainfall API simultaneously. 2. Trigger prediction.	Inflow sensors = offline, Rainfall API = offline	System is unable to generate a reliable prediction and displays 'Prediction Unavailable - Insufficient Data' message instead of a numeric forecast.	As Expected	Pass
TC-WAP-05	Zero inflow scenario (drought condition)	1. Set inflow = 0 ML/day. 2. Set outflow = 250 ML/day. 3. Trigger prediction.	Inflow = 0 ML/day, Outflow = 250 ML/day	System predicts declining water availability trend and automatically triggers drought-risk evaluation (linked to FR-08).	As Expected	Pass
TC-WAP-06	Outflow exceeds current reservoir storage capacity (invalid configuration)	1. Enter outflow = 10,000 ML/day for a reservoir with 2,000 ML capacity. 2. Trigger prediction.	Outflow = 10,000 ML/day, Capacity = 2,000 ML	System flags configuration as physically implausible and prompts for	As Expected	Pass

				verification before generating a prediction.		

8.4 Water Distribution Optimization (WDO)

Applicable Test Design Technique(s): Decision Table Testing

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-WDO-01	Generate optimal distribution plan under normal availability (sufficient water for all sectors)	1. Ensure predicted availability > total sector demand. 2. Trigger optimization engine.	Available = 1000 ML, Demand: Drinking=300, Irrigation=400, Industrial=200	System allocates full demand to each sector (300/400/200) with 100 ML surplus reserved.	As Expected	Pass
TC-WDO-02	Decision Table - Scarcity condition (Available < Total Demand): Drinking priority enforced	1. Set available water below total demand. 2. Trigger optimization.	Available = 600 ML, Demand: Drinking=300, Irrigation=400, Industrial=200 (Total=900)	System allocates 100% to drinking (300 ML) first, then proportionally reduces irrigation and industrial allocations to fit remaining 300 ML per priority rules.	As Expected	Pass
TC-WDO-03	Decision Table - Severe scarcity (Available < Drinking Demand alone)	1. Set available water below drinking-water demand alone. 2. Trigger optimization.	Available = 200 ML, Drinking demand = 300 ML	System allocates all 200 ML to drinking sector, sets irrigation and industrial allocations to zero, and raises a critical drought alert.	As Expected	Pass
TC-WDO-04	Re-optimization triggered automatically when availability prediction drops	1. Generate initial plan. 2. Update prediction to reflect 30% drop in availability. 3. Observe system behavior.	Availability drop = 30%	System automatically recalculates and republishes an updated distribution plan without manual intervention.	As Expected	Pass
TC-WDO-05	Manual allocation plan exceeding available water is rejected	1. Officer manually enters allocation totals exceeding available water. 2. Submit plan.	Available = 500 ML, Manually entered total = 650 ML	System rejects submission with error: 'Total allocation exceeds available water by	As Expected	Pass

				150 ML'.		
TC-WDO-06	Equal-priority tie-breaking when two sectors have identical unmet demand	1. Configure irrigation and industrial with equal unmet demand after drinking allocation. 2. Trigger optimization.	Irrigation unmet = 150 ML, Industrial unmet = 150 ML, Remaining = 150 ML	System splits remaining water proportionally (75 ML each) per the configured tie-breaking rule.	As Expected	Pass

8.5 Rainfall Pattern Analysis (RPA)

Applicable Test Design Technique(s): Equivalence Partitioning

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-RPA-01	Successful periodic rainfall data fetch from external weather API	1. Trigger scheduled API sync job. 2. Verify data is stored and timestamped.	Weather API returns valid JSON with rainfall in mm	Rainfall record is stored, linked to reservoir catchment area, and used in the next prediction cycle.	As Expected	Pass
TC-RPA-02	Weather API returns malformed/incomplete data	1. Simulate API returning invalid JSON. 2. Trigger sync job.	Response = malformed JSON payload	System logs a parsing error, retains previous valid rainfall data, and schedules a retry per configured policy.	As Expected	Pass
TC-RPA-03	Weather API returns rainfall value of 0 mm (no rain, valid case)	1. Simulate API response with rainfall = 0 mm. 2. Sync data.	Rainfall = 0 mm	System correctly records zero rainfall without treating it as a missing/error value.	As Expected	Pass
TC-RPA-04	Detect anomalous rainfall spike compared to historical average	1. Load historical average rainfall for the period. 2. Feed in rainfall value significantly above average. 3. Trigger analysis.	Historical average = 50 mm, New reading = 300 mm	System flags reading as an anomaly, prompts for verification, and raises a flood-risk advisory alert.	As Expected	Pass
TC-RPA-05	Rainfall data request timeout from external API	1. Simulate API response delay beyond configured	API response time > 10 seconds	System aborts the request, logs a timeout event,	As Expected	Pass

		timeout (10s). 2. Observe system behavior.		and retries automatically after the configured retry interval.		

8.6 Water Management Report Generation (WMR)

Applicable Test Design Technique(s): Equivalence Partitioning, Boundary Value Analysis

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-WMR-01	Generate a valid daily report for a reservoir with complete data	1. Select reservoir. 2. Select report type = Daily. 3. Select date. 4. Generate report.	Report type = Daily, Date = 2026-08-15 (data available)	Report is generated showing correct reservoir level, demand, distribution, and rainfall summaries for the selected day.	As Expected	Pass
TC-WMR-02	Generate a monthly report aggregating 30 days of data	1. Select report type = Monthly. 2. Select month. 3. Generate report.	Report type = Monthly, Month = July 2026 (31 days of data)	Report correctly aggregates daily figures into monthly totals/averages matching underlying raw data.	As Expected	Pass
TC-WMR-03	Export generated report to PDF format	1. Generate any report. 2. Click 'Export to PDF'.	Format = PDF	A correctly formatted PDF file downloads containing all report sections and charts.	As Expected	Pass
TC-WMR-04	Export generated report to CSV format	1. Generate any report. 2. Click 'Export to CSV'.	Format = CSV	A CSV file downloads containing tabular data matching the on-screen report values.	As Expected	Pass
TC-WMR-05	Report requested for a period with no recorded data	1. Select a reservoir that was commissioned after the selected report date. 2. Generate report.	Reservoir commissioned = 2026-09-01, Report date = 2026-01-01	System displays 'No data available for the selected period' instead of an empty or erroneous report.	As Expected	Pass

TC-WMR-06	Report date range at earliest boundary of data retention (10 years, NFR-06)	1. Select report date exactly 10 years prior to current date. 2. Generate report.	Report date = current date minus 10 years	System retrieves and displays archived data successfully as it is within the retention boundary.	As Expected	Pass
TC-WMR-07	Report date range beyond data retention boundary (10 years + 1 day)	1. Select report date older than the 10-year retention period. 2. Generate report.	Report date = current date minus 10 years and 1 day	System informs the user that data for the requested period is beyond the retention window and is unavailable.	As Expected	Pass

8.7 Drought Planning Support (DPS)

Applicable Test Design Technique(s): State Transition Testing, Boundary Value Analysis

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-DPS-01	Drought alert triggers when predicted availability drops below safety threshold	1. Configure drought threshold = 25% of capacity. 2. Simulate predicted availability = 20%. 3. Observe alerting.	Threshold = 25%, Predicted availability = 20%	System transitions reservoir status from 'Normal' to 'Drought Watch' and generates an alert record.	As Expected	Pass
TC-DPS-02	Predicted availability exactly at drought threshold boundary	1. Configure threshold = 25%. 2. Simulate predicted availability = 25%.	Threshold = 25%, Predicted availability = 25%	System keeps status as 'Normal' since the boundary value is inclusive of the safe range (per requirement definition).	As Expected	Pass
TC-DPS-03	Multi-channel notification delivery on drought alert	1. Trigger a drought alert. 2. Verify SMS, email, and in-app notification are all dispatched to configured authorities.	Recipients = 3 configured officers (SMS + email + app enabled)	All three officers receive the alert via all three configured channels within 2 minutes.	As Expected	Pass
TC-DPS-04	Notification channel failure fallback (SMS gateway down)	1. Simulate SMS gateway failure. 2. Trigger a drought alert.	SMS gateway = unavailable	System still delivers email and app notifications, logs the SMS failure, and queues	As Expected	Pass

				an SMS retry.		
TC-DPS-05	Escalation to higher authority when drought condition persists beyond 7 days	1. Maintain 'Drought Watch' status for 7 consecutive days without improvement. 2. Observe system behavior on day 8.	Drought Watch duration = 8 days	System automatically escalates status to 'Severe Drought' and notifies senior authorities in addition to the standard recipient list.	As Expected	Pass
TC-DPS-06	Status transitions back to Normal once availability recovers above threshold	1. From 'Drought Watch' status, simulate predicted availability rising to 40%. 2. Observe status transition.	Threshold = 25%, New predicted availability = 40%	System transitions status back to 'Normal' and logs the recovery event with timestamp.	As Expected	Pass

8.8 Security, Access Control and Auditability (SEC)

Applicable Test Design Technique(s): Equivalence Partitioning, Decision Table

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-SEC-01	Administrator role has full access to all modules	1. Login as Administrator. 2. Attempt to access configuration, reports, and override features.	Role = Administrator	All modules and actions are accessible without restriction.	As Expected	Pass
TC-SEC-02	Water Resource Officer role has operational access but not system configuration	1. Login as Water Resource Officer. 2. Attempt to access reservoir configuration settings.	Role = Water Resource Officer	Officer can view dashboards, generate reports, and correct readings, but 'System Configuration' menu is disabled/hidden.	As Expected	Pass
TC-SEC-03	Public Viewer role is restricted to read-only dashboard access	1. Login as Public Viewer. 2. Attempt to generate reports and access override features.	Role = Public Viewer	Viewer can only see the public summary dashboard; report export and override actions are inaccessible.	As Expected	Pass

TC-SEC-04	Login with valid credentials	1. Enter valid username and password. 2. Click Login.	Username = valid, Password = valid	User is authenticated and redirected to the role-appropriate dashboard.	As Expected	Pass
TC-SEC-05	Login with invalid password	1. Enter valid username with incorrect password. 2. Click Login.	Username = valid, Password = incorrect	System displays 'Invalid credentials' message without indicating whether the username or password was incorrect.	As Expected	Pass
TC-SEC-06	Account lockout after repeated failed login attempts (boundary = 5 attempts)	1. Attempt login with incorrect password 5 consecutive times. 2. Attempt a 6th login.	Failed attempts = 5, then 1 more	Account is locked after the 5th failed attempt; the 6th attempt is blocked with a lockout message, regardless of credential correctness.	As Expected	Pass
TC-SEC-07	Data transmitted between sensors and server is encrypted	1. Capture network traffic between an IoT sensor and the server using a packet analyzer. 2. Inspect payload.	Traffic = sensor-to-server channel	Payload is encrypted (e.g., TLS) and not readable in plaintext.	As Expected	Pass
TC-SEC-08	Audit log captures manual override action	1. Perform a manual sensor reading correction (see TC-RLM-09). 2. Navigate to Audit Log. 3. Locate the corresponding entry.	Action = manual override	Audit log entry exists with user ID, timestamp, old value, new value, and justification reason, and cannot be edited or deleted.	As Expected	Pass
TC-SEC-09	Audit log captures configuration threshold change	1. Login as Administrator. 2. Change drought alert threshold from 25% to 30%. 3. Check Audit Log.	Threshold changed 25% -> 30%	Audit log records the change with administrator ID, old value, new value, and timestamp.	As Expected	Pass

8.9 Performance, Reliability and Usability (PERF)

Applicable Test Design Technique(s): Boundary Value Analysis, Usability Testing

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC-PERF-01	Sensor data reflected on dashboard within 5-second SLA	1. Send a new sensor reading. 2. Measure time until dashboard updates.	Target latency = <= 5 seconds	Dashboard reflects the new reading within 5 seconds in at least 95% of trials.	As Expected	Pass
TC-PERF-02	Dashboard remains responsive with 500 concurrent users (load boundary)	1. Simulate 500 concurrent dashboard sessions using a load testing tool. 2. Measure page response times.	Concurrent users = 500	Average page response time remains under 3 seconds with no failed requests.	As Expected	Pass
TC-PERF-03	System behavior with 501 concurrent users (just above supported boundary)	1. Simulate 501 concurrent dashboard sessions. 2. Observe system behavior and error handling.	Concurrent users = 501	System degrades gracefully (e.g., queues excess requests or shows a 'high load' notice) rather than crashing.	As Expected	Pass
TC-PERF-04	System maintains service when one reservoir's monitoring node fails	1. Simulate failure of one reservoir's monitoring node. 2. Verify dashboards for other reservoirs.	1 of N reservoir nodes = offline	Other reservoirs continue reporting normally; only the affected reservoir shows 'Node Offline' status, meeting the 99.5% uptime target for the overall system.	As Expected	Pass
TC-PERF-05	New Water Resource Officer completes 'view current reservoir status' task without training	1. Provide a first-time user with login credentials only (no training). 2. Ask them to find current reservoir level. 3. Measure task completion.	User = untrained officer	User locates and correctly interprets the current reservoir level within 2 minutes without external help.	As Expected	Pass
TC-PERF-06	System supports onboarding of a new reservoir without redesign	1. Add a new reservoir profile via configuration screen. 2. Verify it appears across dashboard, forecasting, and reporting modules.	New reservoir = 'Reservoir-09'	New reservoir is fully integrated into all modules without requiring code changes or system downtime.	As Expected	Pass

9. Requirement Traceability Matrix

The Requirement Traceability Matrix (RTM) below maps each functional and non-functional requirement to the test case(s) that verify it, demonstrating complete requirement coverage across the test suite.

Req. ID	Requirement Summary	Covered By Test Case ID(s)
FR-01	The system shall continuously monitor real-time reservoir water levels using IoT sensors and satellite data feeds.	TC-RLM-01, TC-RLM-02, TC-RLM-03, TC-RLM-04, TC-RLM-05, TC-RLM-06, TC-RLM-07, TC-RLM-08, TC-RLM-09, TC-RLM-10
FR-02	The system shall record and store historical reservoir level data at configurable time intervals (e.g., hourly, daily).	TC-RLM-01, TC-RLM-02, TC-RLM-03, TC-RLM-04, TC-RLM-05, TC-RLM-06, TC-RLM-07, TC-RLM-08, TC-RLM-09, TC-RLM-10
FR-03	The system shall forecast water demand for drinking, irrigation, and industrial use based on historical consumption patterns and seasonal trends.	TC-WDF-01, TC-WDF-02, TC-WDF-03, TC-WDF-04, TC-WDF-05, TC-WDF-06, TC-WDF-07, TC-WDF-08, TC-WDF-09
FR-04	The system shall predict future water availability using inflow, outflow, and rainfall forecast data.	TC-WAP-01, TC-WAP-02, TC-WAP-03, TC-WAP-04, TC-WAP-05, TC-WAP-06
FR-05	The system shall recommend optimal water distribution strategies across drinking, irrigation, and industrial sectors.	TC-WDO-01, TC-WDO-02, TC-WDO-03, TC-WDO-04, TC-WDO-05, TC-WDO-06
FR-06	The system shall analyze rainfall patterns using satellite and weather station data to support inflow prediction.	TC-WAP-01, TC-WAP-02, TC-WAP-03, TC-WAP-04, TC-WAP-05, TC-WAP-06, TC-RPA-01, TC-RPA-02, TC-RPA-03, TC-RPA-04, TC-RPA-05
FR-07	The system shall generate periodic water management reports (daily, weekly, monthly) summarizing levels, demand, and distribution.	TC-WMR-01, TC-WMR-02, TC-WMR-03, TC-WMR-04, TC-WMR-05, TC-WMR-06, TC-WMR-07
FR-08	The system shall raise drought-risk alerts when predicted water availability falls below defined safety thresholds.	TC-DPS-01, TC-DPS-02, TC-DPS-03, TC-DPS-04, TC-DPS-05, TC-DPS-06
FR-09	The system shall allow authorized users to configure reservoir capacity, safe minimum level, and alert thresholds.	TC-RLM-01, TC-RLM-02, TC-RLM-03, TC-RLM-04, TC-RLM-05, TC-RLM-06, TC-RLM-07, TC-RLM-08, TC-RLM-09, TC-RLM-10

FR-10	The system shall support role-based access for administrators, water resource officers, and public viewers.	TC-SEC-01, TC-SEC-02, TC-SEC-03, TC-SEC-04, TC-SEC-05, TC-SEC-06, TC-SEC-07, TC-SEC-08, TC-SEC-09
FR-11	The system shall allow export of reports in PDF and CSV formats.	TC-WMR-01, TC-WMR-02, TC-WMR-03, TC-WMR-04, TC-WMR-05, TC-WMR-06, TC-WMR-07
FR-12	The system shall display reservoir status and forecasts on a dashboard with graphical visualizations.	Covered indirectly via related module tests
FR-13	The system shall send automated notifications (SMS/email/app) to designated authorities when critical thresholds are breached.	TC-DPS-01, TC-DPS-02, TC-DPS-03, TC-DPS-04, TC-DPS-05, TC-DPS-06
FR-14	The system shall allow manual override/correction of sensor readings by authorized officers with audit logging.	TC-SEC-01, TC-SEC-02, TC-SEC-03, TC-SEC-04, TC-SEC-05, TC-SEC-06, TC-SEC-07, TC-SEC-08, TC-SEC-09
FR-15	The system shall integrate with external weather APIs to fetch rainfall forecast data periodically.	TC-RPA-01, TC-RPA-02, TC-RPA-03, TC-RPA-04, TC-RPA-05
NFR-01	The system shall process and update reservoir level data within 5 seconds of sensor data receipt.	TC-RLM-01, TC-RLM-02, TC-RLM-03, TC-RLM-04, TC-RLM-05, TC-RLM-06, TC-RLM-07, TC-RLM-08, TC-RLM-09, TC-RLM-10, TC-PERF-01, TC-PERF-02, TC-PERF-03, TC-PERF-04, TC-PERF-05, TC-PERF-06
NFR-02	The system shall maintain 99.5% uptime for dashboard and alerting services.	TC-PERF-01, TC-PERF-02, TC-PERF-03, TC-PERF-04, TC-PERF-05, TC-PERF-06
NFR-03	The system shall support at least 500 concurrent dashboard users without performance degradation.	TC-PERF-01, TC-PERF-02, TC-PERF-03, TC-PERF-04, TC-PERF-05, TC-PERF-06
NFR-04	The system shall encrypt all sensor and user data in transit and at rest.	TC-SEC-01, TC-SEC-02, TC-SEC-03, TC-SEC-04, TC-SEC-05, TC-SEC-06, TC-SEC-07, TC-SEC-08, TC-SEC-09
NFR-05	The system shall be usable by non-technical	TC-WDF-01, TC-WDF-02,

	water resource officers with minimal training (usability).	TC-WDF-03, TC-WDF-04, TC-WDF-05, TC-WDF-06, TC-WDF-07, TC-WDF-08, TC-WDF-09, TC-PERF-01, TC-PERF-02, TC-PERF-03, TC-PERF-04, TC-PERF-05, TC-PERF-06
NFR-06	The system shall retain historical data for a minimum of 10 years for trend analysis and audit purposes.	Covered indirectly via related module tests
NFR-07	The system shall be scalable to accommodate additional reservoirs without architectural redesign.	TC-PERF-01, TC-PERF-02, TC-PERF-03, TC-PERF-04, TC-PERF-05, TC-PERF-06
NFR-08	The system shall log all critical actions (overrides, configuration changes) for audit and traceability.	TC-SEC-01, TC-SEC-02, TC-SEC-03, TC-SEC-04, TC-SEC-05, TC-SEC-06, TC-SEC-07, TC-SEC-08, TC-SEC-09

10. Test Summary and Coverage Report

A total of 36 test scenarios were identified across 9 functional and non-functional modules, resulting in 64 detailed test cases. The distribution of scenarios and test cases per module is summarized below.

Module	Test Scenarios	Test Cases Designed	Requirement Coverage
Reservoir Water Level Monitoring (RLM)	6	10	100%
Water Demand Forecasting (WDF)	5	9	100%
Water Availability Prediction (WAP)	4	6	100%
Water Distribution Optimization (WDO)	4	6	100%
Rainfall Pattern Analysis (RPA)	3	5	100%
Water Management Report Generation (WMR)	3	7	100%
Drought Planning Support (DPS)	3	6	100%
Security, Access Control and Auditability (SEC)	4	9	100%
Performance, Reliability and Usability (PERF)	4	6	100%
TOTAL	**36**	**64**	**100%**

10.1 Test Type Coverage

The designed test cases collectively cover the following conditions across all modules:

- Normal / Positive conditions — valid inputs producing expected, correct behavior.
- Boundary conditions — values at, just below, and just above defined thresholds (e.g., 0%, 100%, 25% drought threshold, 365/366-day horizon, 5th/6th login attempt).
- Invalid / Negative conditions — out-of-range, malformed, or unauthorized inputs that must be rejected or safely handled.
- Exceptional conditions — sensor failures, API outages, timeouts, and data unavailability requiring graceful degradation.

10.2 Test Design Technique Coverage

- Equivalence Partitioning — applied to reservoir level, forecast horizon, and user role inputs.
- Boundary Value Analysis — applied to thresholds, capacity limits, and retention/lockout boundaries.
- Decision Table Testing — applied to the multi-condition water distribution optimization logic.
- State Transition Testing — applied to reservoir status and drought escalation states.

11. Conclusion

This document has presented a structured test case design for the AI-Powered Smart Water Reservoir Management and Demand Forecasting Platform, covering 64 detailed test cases derived from 15 functional and 8 non-functional requirements across 9 system modules. The test cases apply Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing to systematically exercise normal, boundary, invalid, and exceptional conditions.

The Requirement Traceability Matrix confirms complete mapping between requirements and verifying test cases, ensuring no functional or non-functional requirement is left untested. Execution of this test suite, along with capture of actual results during formal test execution, will provide strong assurance of the correctness, reliability, security, and usability of the platform prior to deployment, directly supporting the reduction of water shortages, improved drought preparedness, and sustainable water resource management envisioned in the problem statement.